

Pratik Bairoliya

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EXPERIENCE

Capital One

Jul. 2026 – Present

Software Engineer II

McLean, VA

- Own the nightly **end-of-day close** — posting, settlement, ledger cutoff across Card and Bank, run on **Temporal**.
- Lead the **REST API** replacing the vendor's close operator tooling, bringing nightly controls in-house.
- Monitoring and distributed tracing in **OpenTelemetry** and **Splunk**: alerts page on-call mid-run, not at cutoff.
- Ship end-of-day services on **Kubernetes** through **Argo CD** and **Helm**, rollback one revision away.

Capital One

Aug. 2025 – Jul. 2026

Software Engineer I

McLean, VA

- Wrote the ledger's **batch write path** in **Go**, empty repo to production; every nightly cycle runs through it.
- Designed a **block-allocation ID service**: one call reserves a range, taking per-ID contention off bulk opens.
- Built the nightly **PySpark/Databricks** reconciliation: day-over-day aggregate diffs alert before a bad cycle ships.
- Cut data egress on the nightly **99M+ Card account** publish by performance-tuning **Hive** sort on **AWS EMR**.
- Rolled **proof-of-possession (PoP) tokens** across Card and Bank endpoints, binding each call to a key a thief lacks.

Capital One

May 2024 – Aug. 2024

Software Engineering Intern

Richmond, VA

- Moved the military rewards catalog off SharePoint into a **Vue.js/Spring Boot/DynamoDB** app: **2.8x** faster.
- Shipped it to production myself on **Terraform**, **Docker** and **Jenkins** pipelines.

Fidelity Investments

Jun. 2023 – May 2024

Software Engineer

Durham, NC

- Migrated Fidelity's mainframe files and job calls to **AWS** with **Terraform**, serving **400+** associates.
- Built the single source of truth for every **mainframe API** as a **Neo4j** graph, cutting **5,000+** duplicate nodes.
- Delivered the full stack on a 4-person team: **Angular** UI, **Java** services, **S3** metadata sync.

PROJECTS

Lookback — automated daily briefing | *Python, Ollama (local LLM), Google APIs, SQLite, launchd* 2026

- Scheduled job merging Calendar, Gmail, and iMessage into one morning note; runs **100% on-device**.
- Safe to re-run**: a missed night rebuilds the same note instead of duplicating it.

Pantry — receipt-to-groceries tracker | *Python, Apple Vision OCR, Ollama, file watchers* 2026

- OCR** ingest pipeline turning receipt photos into a pantry list with spend rolled up by category.
- Idempotent ingest**: rescanning a receipt cannot double-count, so spend totals stay correct.

Unix Shell & OS Internals | *C, C++, CUDA, pthreads, sockets* 2022–24

- A Unix shell with no standard library, a generic hash map on hand-written vtables, and a threaded Scrabble server.
- Max-subarray solved five ways — fork/pipe, threads, semaphores, monitors, then **CUDA** at **1.7s** on an RTX 2070.
- One puzzle across three **IPC** mechanisms — POSIX message queues, System V shared memory, TCP sockets — with **RSA** challenge/response auth.

Game AI | *C++, SFML* 2024

- Steering that composes by summing accelerations, **A*** with swappable heuristics, and a decision tree on top.

TECHNICAL SKILLS

Languages: Go, Java, Python, TypeScript/JavaScript, SQL, C/C++, Bash

Infrastructure: AWS (EMR, S3, DynamoDB), Kubernetes, Temporal, Docker, Terraform, Argo CD, Helm, CI/CD, Linux

Backend & Data: REST APIs, microservices, distributed batch, ETL, Spark (PySpark), Hive, Databricks, Spring Boot

Frontend & Other: Vue.js, Angular, Node.js, local LLM inference (Ollama), observability (Splunk, OpenTelemetry)

Certification: AWS Certified Solutions Architect – Associate (Mar. 2026)

EDUCATION

North Carolina State University

Aug. 2021 – May 2025

B.S. Computer Science and B.S. Applied Mathematics — GPA 4.0

Raleigh, NC

- Park Scholar** — full merit scholarship to the top **1%** of applicants for leadership and service.
- Teaching Assistant** — CSC 221 (Python) and MA 405 (Linear Algebra): labs, office hours, grading.
- Undergraduate Researcher**, 2022–2024 — built **Python** tooling that automated a faculty group's data analysis.
- Interests** — competitive Bhangra dance, Taekwondo, weightlifting, and the Krispy Kreme Challenge.
- Co-founder, InspireNC** — FIRST Robotics 6908 for **100+** students; taught **Java** to cohorts of 35.